2. They are stable
3. They are used either repeatedly in a series of batch wise reactions or continuously in flow systems.

HOTS

The technique of immobilized enzymes may increase the use of enzymes in industry for product modification. Why? Give reason.

Summary

Industrial microbiology is a branch, of microbiology that deals with the study and uses of various microorganisms. The birth of industrial microbiology largely began with the studies of Pasteur on fermentation. Various products of both primary and secondary metabolites are produced by different microorganisms. Both primary and secondary screening is involved in the isolation of industrially important microorganisms. The isolated strains are modified for higher yield through various procedures (example) protoplast fusion. Thus strain improvement improves the fermentation efficiency. Fermentation is carried under a suitable parameters, in a controlled environment is called Fermentor. Fermentation process involves both cup stream processing and downstream processing. In cup stream processing, inoculum preparation scale up, preparation of medium and sterilization of media, are carried out.

Down stream processing, includes the separation and purification of products. The industrially important products like penicillin, wine and citric acid, single cell protein, are produced in large scale by using microorganisms.

Immobilization is need for the physical or chemical fixation of plant, animal cells, organelles, enzymes or other proteins into a solid matrix or retained by a membrane. Immobilized cells can be used for repeated doses.

Evaluation

Multiple choice questions

M8VD89

1. The term fermentation originates from a Latin verb _____
   a. Wear
   b. Fervere
   c. Severe
   d. Cheer
2. _____ metabolites are produced in small quantities during industrial production.
   a. Secondary metabolites
   b. Primary metabolites
   c. Tertiary metabolites
   d. Neutral metabolites
3. _____ enrichment technique for isolation of secondary metabolite producing Actinomycetes.
   a. Soil-Sodium Carbonate
   b. Soil-Calcium Carbonate
   c. Soil-Strontium Carbonate
   d. Soil-Potassium Carbonate
4. The microbes used in the industrial microbiology has these qualities
   Statement A: The strain should have stable biochemical and genetical characteristics.

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